Homework 1: Camera Capture, Camera Calibration and Filtering

50 points possible

You may use the OpenCV or MATLAB to complete these exercises. Please choose one of these libraries and plan on using it throughout the course. Many of the tools developed here will be useful in upcoming assignments.

1. Create a program that will read in all the images from a folder and use them to get the calibration parameters. Calibrate your camera using a minimum of three data sets. Compare and contrast the results. Do they make sense. One of the data sets should have at least 25 images. Calibrate the camera using 2, 5, 10, 15 and 25 images. Compare the results. What do they tell us?
2. Filter the noisy images using a 5x5 gaussian kernel, a 3x3 gaussian kernel and another noise suppressing filter. Compare and contrast the results. Do this with the images listed below:
   - Camera_man_w_noise
   - SaltAndPepperNoise
   - GausianNoise
   - UniformNoise

   Make sure to answer questions like which image each filter works best for? What metric is used to determine the quality of the filtering? What filter would be recommended for what types of noise? The writeup needs to contain filtered and original images. It also needs to describe the code, what functions were used, inputs to the function, and settings. The selection of setting and such need to be justified in your writeup.

3. Repeat the above with a median filter and an n-rank filter. Use both a 3x3 and 5x5 window.
4. Use the thresholding tutorial provided on https://docs.opencv.org/4.x/d7/d4d/tutorial_py_thresholding.html. Perform both simple and adaptive thresholding. Compare both methods and discuss the tradeoffs between them. Make sure to include multiple images demonstrating the findings discussed in the writeup.
   - Coins1
   - Coins2
   - screws
5. Use morphological operators to improve the segmentation of the images in problem 3. There are multiple operations with different masks and orders. Describe which set of operations works best and why. Make sure to include all images generated. See https://docs.opencv.org/4.x/d9/d61/tutorial_py_morphological_ops.html for help on how to use the operators.

The write-ups must contain enough detail that I could hand it to any student in the class and they would be able to repeat the work. In addition, all mathematical equations must be provided, and their meaning explained. Write-ups that do not make it clear how each step was completed will receive little, if any credit. The code used should be provided with the writeup.